WEEK 4 RESEARCH SYNTHESIS · ALPHAEARTH

Residential Signal in Frozen Embeddings

Compact, partially shared, and geographically brittle across urban metros.

64-D Frozen AE

10 Metros · 1.75M Cells

Leave-One-Region-Out (LORO)

BENCHMARK RIGOR

Standardized Experimental Protocol

10 Metros

Downtown core extraction boxes.

1.75M Cells

Balanced 1:1 RES vs NRES cells.

64-D Latent

Frozen AlphaEarth representations.

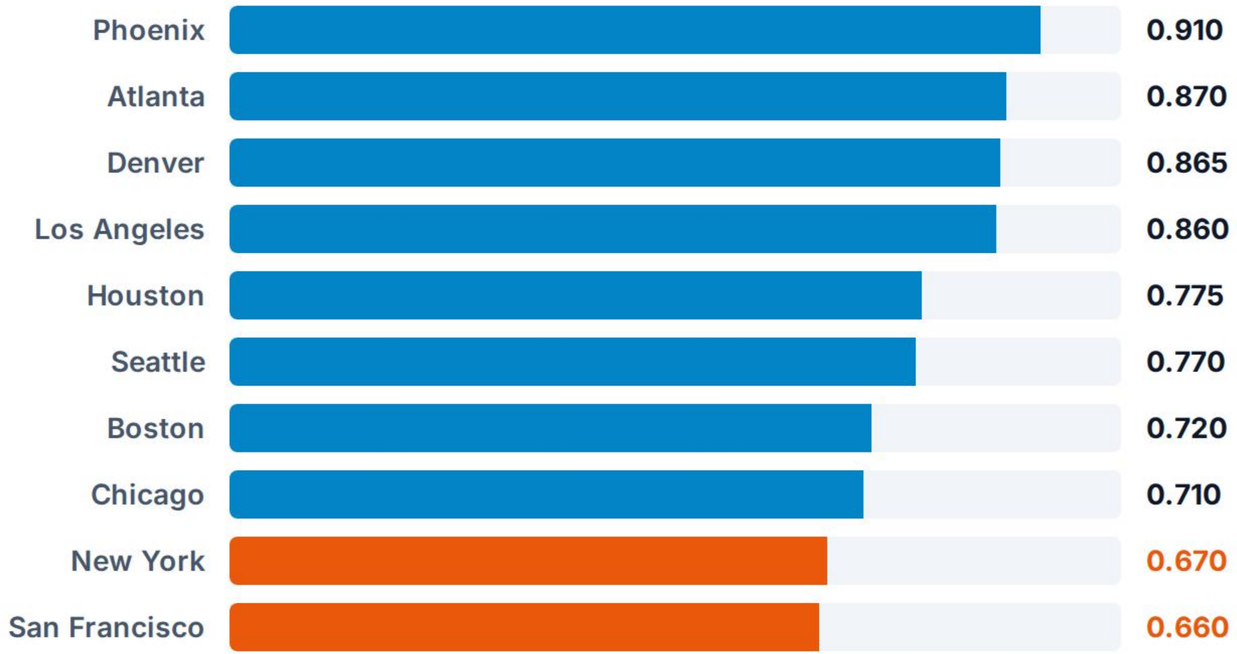
LORO Split

Rigorous hold-out at 0.50 threshold.

Controlled baseline: identical sampling parameters and metrics applied across all compression runs.

PART I · LINEAR BASELINES

Mean LORO F1: 0.779 (Metro Disparity: ±0.089)



TOP PERFORMER

Phoenix: 0.910

Suburban roof footprints clearly isolate.

URBAN FAILURE ZONE

NYC & SF: ~0.66

Dense mixed-use buildings collapse signal.

DIMENSIONALITY REDUCTION

Building Signal Sits in a Short Supervised Map

SUPERVISED PLS

k = 4

Captures ≥95% of full 64-D F1 score.

Supervised direction extraction isolates functional variance immediately.

UNSUPERVISED PCA

k = 15

Required to reach the same 95% retention.

Dominant variance axes trace geographic differences rather than function.

L1 REGULARIZATION

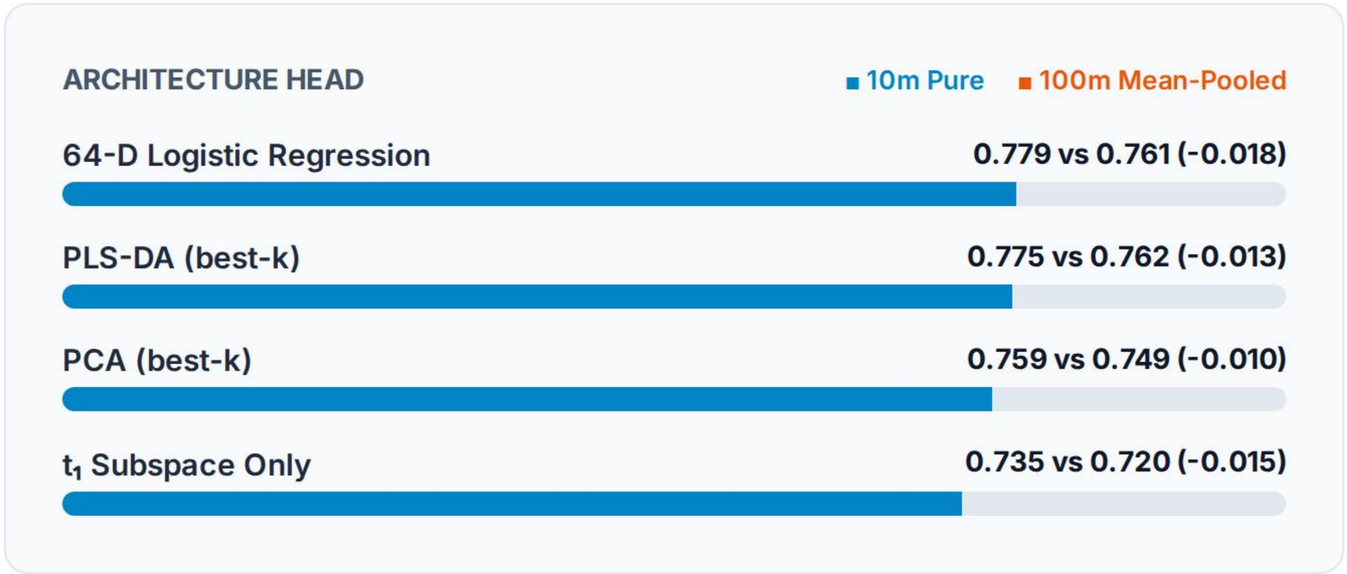
t_1 Axis

Head collapses entirely onto first component.

At C=10, LORO F1 achieves 0.735 using almost exclusively component t_1.

SPATIAL AGGREGATION

Mean-Pooling to 100m Costs Only 0.01–0.02 F1



Coarse aggregation preserves dominant direction

Even after a 10× spatial blur, the primary residential projection axis remains robust with minimal degradation.

PART II · ARCHITECTURE COMPARISON

Complex Deep Architectures Fail to Beat Linear Models

Model / Run	LORO F1	Mean Held-Out Gap
100m Mean-Pool Linear (LR)	**0.761**	—
ViT 128-D	**0.756**	—
v4 Within-Region Gap	**0.738**	2.13
v3 Longer Training	**0.735**	1.71
v1 BCE + SupCon	**0.732**	1.76

Linear Head Remains Optimal

Deep non-linear feature maps overfit to training metro specifics.

Contrastive Loss Overfits

SupCon forces separation on axes that fail under cross-metro transfer.

LATENT SPACE TOPOGRAPHY

Atlanta Clears Gap of 3.51 · NYC Collapses to 0.63

Atlanta

Success Case

Gap: 3.51

LORO F1: 0.898

Residential and commercial clouds fully separate across the decision plane.

New York

Collapse Case

Gap: 0.63

LORO F1: 0.586

Residential and non-residential form a single indistinguishable cloud.

Phoenix

Validation

Gap: 5.44

Held-Out Separability

Maintains extreme separation while NYC is held out, proving localized failure.

PART III · GEOGRAPHIC TRANSFER

Key Empirical Answers Across the Suite

Signal in Frozen Embeddings?	**0.779 F1**	Linear heads extract meaningful building signal without fine-tuning.
Subspace Intrinsic Dimension?	**k = 4 (PLS)**	Supervised mapping isolates signal far faster than PCA (k=15).
Subspace Transferability?	**Partly Shared**	Low-dimensional PLS heads transfer substantially better than full 64-D LR.

Core takeaway: full 64-D vectors carry spurious regional markers that damage generalization.

CROSS-METRO MATRIX

Short PLS Discards Metro-Specific Artifacts

PLS (k=2) Transfer

Compressing to 2 dimensions forces the model to ignore local geographic traits. Cross-metro retention improves significantly.

Full 64-D Vulnerability

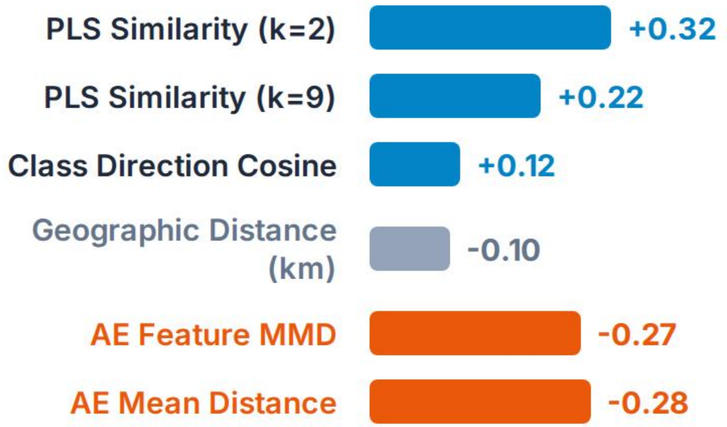
Full 64-D vectors encode localized climate, vegetation, and soil features. Classifiers anchor on these spurious indicators.

Net Transfer Gain

ΔT (PLS-2 minus 64-D) delivers up to +0.40 retention improvements on historically difficult pairs like Boston and Denver.

SPEARMAN CORRELATION ANALYSIS

Subspace Overlap Drives Transfer · Geography Does Not



Subspace Alignment Dictates Success

The degree of mathematical subspace overlap is the single strongest predictor of transfer retention.

Physical Proximity is Irrelevant

Geographic distance (rho = -0.10) shows negligible impact compared to latent feature shifts.

TRADE-OFF OPTIMIZATION

PLS k=2: Preserves Labels, Purges Metro Identity

THE DESIRED PARETO CORNER

High F1 Score + Low Metro Accuracy

An ideal representation preserves building function while completely erasing which city the cell originated from.

PLS-2 F1 Score	Metro Accuracy
0.775	**~0.30**

PCA Leaks Metro Identity

At similar F1 levels (k=16), PCA achieves >0.95 metro identity, clinging to local geography.

Full 64-D Over-Identifies

Full AE embeddings yield near 1.0 metro accuracy, guaranteeing domain overfitting.

STRATEGIC CONCLUSIONS

What Holds vs. What Fails

What Holds

- ✓ **~0.78 LORO F1:** Solid signal in frozen AE.
- ✓ **Short Subspace:** PLS k=2–4 retains signal.
- ✓ **Better Transfer:** Low-D heads ship more robustly.

What Does Not Hold

- ✗ **Global Transfer:** Dense urban cores collapse.
- ✗ **Complex Tile Encoders:** Deep models overfit.
- ✗ **Geography as Driver:** Physical distance does not predict.

> **Guiding Principle: Retain only the short residential projection direction — geography lives in the rest.**